QP Code: 25900042

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-BCA (Honours) REGULAR EXAMINATION MARCH 2025
SECOND SEMESTER
Core Course (CC) – MG2CCRBCA101 – DATA STRUCTURES
(2024 Admission Onwards)

Answer Key

Duration: 2 Hours Maximum Marks: 70

Part A

Very Short Answer Questions. Answer all questions.

Each question carries 2 marks. [2 × 5 = 10]

1. **Explain the memory representation of a linear array.** [U] / [CO1]
   – Definition of linear array (1 mark)
   – Elements are stored in consecutive memory locations (1 mark)

2. **Explain the role of a stack when evaluating a postfix expression. Provide an example to illustrate how the stack is used.** [U] / [CO2]
   – A stack is used to evaluate postfix (Reverse Polish) expressions efficiently. The stack helps to manage operands and intermediate results. (1 mark)

   -- Example: 5 6 + 2 * → (5 + 6) * 2 = 22 with stack operations (1 mark)

3. **Describe how a singly linked list is represented in memory.** [U] / [CO3]

   – Explanation of node structure with data and next (1 mark)
   – Use of pointers for sequential linking (1 mark)

4. **Describe connected graphs using a suitable example.** [U] / [CO4]
   – Definition of connected graph (1 mark)
   – Example: with vertices and edges (1 mark)

5. **Distinguish a priority queue and a normal queue.** [An] / [CO2]

   – Difference in operation: FIFO vs priority-based processing (1 mark)
   – Real-life example: hospital queue (1 mark)

Part B

Short Answer Questions. Answer any 5 questions.
Each question carries 6 marks. [6 × 5 = 30]

6. **Describe different types of time complexity analysis with suitable examples.**

[U] / [CO1]

– Best, worst, average case definitions (3 marks)
– Examples like Linear Search, Binary Search (3 marks)

7. **Write the algorithm for the "push" operation in a stack. Discuss the limitations of using arrays for implementing stacks.** [An] / [CO2]
   – Stack push algorithm steps (3 marks)
   – Limitations: fixed size, memory wastage, overflow (3 marks)

8. **Distinguish between closed hashing and open hashing.** [An] / [CO3]

   – Definition of open hashing (2 marks)
   – Definition of closed hashing (2 marks)
   – Differences (2 marks)

9. **Compare binary trees and binary search trees with examples.** [An] / [CO4]

   – Definitions of binary tree and BST (2 marks)
   – Examples (2 marks)
   – Key differences (2 marks)

10. **Explain the Insertion Sort algorithm. Describe its working with an example.**

[U] / [CO1]

– Algorithm steps or program (3 marks)
– Step-by-step working with an example (3 marks)

11. **Write an algorithm or a program to traverse a singly linked list and print all its elements.** [A] / [CO3]
    – Proper algorithm or program (3 marks)
    – Explanation (3 marks)

12. **Analyze the advantages and disadvantages of using an AVL tree over a standard Binary Search Tree.** [An] / [CO4]
    – AVL advantages: balanced height, O(log n) operations, Efficient Searching (3 marks)
    – Disadvantages: complex rotations, overhead (3 marks)

Part C

Essay Questions. Answer any 2 questions.
Each question carries 15 marks. [15 × 2 = 30]

13. **Explain two dimensional arrays in C. Write a C program to find the row sum of a matrix.** [A] / [CO1]
    – Definition and memory layout of 2D array with explanation (7 marks)
    – C program to calculate row sum (8 marks)

14. **Explain the array implementation of a simple queue and circular queue with algorithms and examples.** [An] / [CO2]

    – Structure and operations of a simple queue with example (7 marks)
    – Structure and operations of a circular queue with example (8 marks)

    Explain insert, delete and traverse operations with algorithm/code.

15. **Define recursion. What are the applications of recursion? Discuss the significance of run time stack in recursive program implementation.** [An] / [CO3]

    – Definition and working of recursion (4 marks)
    – Applications: factorial, Fibonacci, etc.. Explain any one application with example. (5 marks)
    – Runtime stack: call stack, frames, example (6 marks)